

# Parallel Hybrid Retrieval: Enhancing Temporal RAG via Concurrent Graphiti and Semantic Search

## 1. Problem Statement

Standard hybrid Retrieval-Augmented Generation (RAG) architectures often fail to account for complex temporal and historical changes within large document corpora. Consequently, models suffer from temporal hallucinations. Furthermore, querying multiple complex indexes (like vector stores and knowledge graphs) sequentially creates significant inference bottlenecks.

## 2. Motivation

Using traditional Semantic Search over chunked documents (RAG) suffers from poor relational knowledge that Knowledge Graphs aim to solve. By merging them, you get a system that can find the right document (Semantic RAG) while ensuring the AI doesn't use outdated information from that document (Graphiti), ultimately solving the "hallucination of state" where an AI remembers a fact but forgets that it is no longer true.

## 3. Objectives

- To implement a dual-retrieval RAG architecture combining semantic search and Graphiti.
- To parallelize the retrieval processes to reduce overall inference latency.
- To evaluate the architecture's groundedness and faithfulness against standard sequential models.

## 4. Technical Contributions

**Architectural Improvement:** A novel hybrid approach that fuses standard chunked semantic search with temporal knowledge graphs to accurately track historical state changes in text.

## 5. Dataset(s)

- **Evaluation Dataset:** RAGTruth (for hallucination and faithfulness validation).

## 6. Baseline Papers / Methods

- **Method 1:** Standard Hybrid RAG (Reference: arXiv:2408.04948).
- **Method 2:** Temporal Knowledge Graphs (Graphiti) (Reference: arXiv:2501.13956)

## 7. Proposed Methodology / Flow

1. **Ingestion & Indexing:** Documents are chunked and ingested into a Vector Database and Graphiti.
2. **Parallel Retrieval (PDC Integration):** Upon receiving a query, the system spawns parallel processes. Process A executes the semantic vector search. Process B concurrently queries the Graphiti temporal graph.
3. **Aggregation:** Results from both processes are joined, deduplicated, and ranked.
4. **Generation & Evaluation:** The aggregated context is passed to the LLM. The output is evaluated using TruLens and RAGTruth for groundedness, context relevance, and faithfulness.

## 8. Expected Outcomes

- A measurable decrease in temporal hallucinations and an increase in groundedness and faithfulness scores compared to standard baselines.